Arabic Fake News Detection in Egypt NLP
Phase 1 and Phase 2

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Phase 1 Report: Data Preprocessing and Visualization

1. Introduction

The objective of this phase was to prepare a high-quality dataset for fake news detection in Egyptian Arabic text and explore its characteristics using preprocessing, feature engineering, and visualization techniques. Since Arabic is a morphologically rich language with dialectal variations, effective preprocessing is essential to improve model performance.

2. Dataset Description

The main dataset used in this project was the Egypt Fake Tweets Labeled Dataset, which contains Egyptian Arabic text samples with the following labels:

- r = Real
- f = Fake
- idk = Uncertain / No decision

To ensure label reliability, ambiguous samples labeled **idk** were removed. The final binary classification dataset contained:

- Real: 5,286 samples
- Fake: 1,617 samples
- Total: 6,903 samples

In addition, multiple external Egyptian Arabic corpora (tweets, Reddit, Wikipedia, political text, etc.) containing approximately 519,640 text samples were collected for training custom word embeddings.

3. Text Preprocessing

Several preprocessing steps were applied to normalize and clean Arabic text:

- Lowercasing text
- Removing URLs, mentions, and hashtag symbols
- Removing punctuation and numbers
- Removing Arabic diacritics
- Normalizing letters (e.g. , , $\rightarrow$)
- Converting $\rightarrow$ and $\rightarrow$
- Removing extra spaces
- Removing Arabic stopwords

These steps reduced noise and improved feature consistency.

4. Exploratory Data Analysis

Class Distribution

The dataset showed class imbalance, with significantly more real samples than fake samples:

- Real: 76.6%
- Fake: 23.4%

Text Length Analysis

Most texts were short and tweet-like, ranging between 6 and 14 words, with an average around 9–10 words.

5. Feature Representation

TF-IDF

TF-IDF vectorization was applied using unigram and bigram features. This representation captures the importance of words relative to the dataset and is highly effective for short text classification.

Custom Word2Vec

A Word2Vec model was trained on the external Egyptian Arabic corpora:

- Corpus size: 494,954 tokenized sentences
- Vocabulary size: 93,725 words
- Embedding dimension: 100

This produced domain-specific Egyptian Arabic embeddings.

6. Visualization

PCA

Principal Component Analysis (PCA) was applied to reduce TF-IDF vectors into two dimensions. The resulting plot showed partial separation between fake and real samples.

t-SNE

t-SNE visualization revealed local clusters and confirmed that the classes share overlapping patterns while still maintaining separable regions.

7. Observations

- The dataset is moderately imbalanced.
- Fake and real texts share linguistic similarities.
- TF-IDF features capture useful discriminative patterns.
- Custom embeddings provide richer semantic understanding of Egyptian Arabic.

8. Conclusion

Phase 1 successfully prepared the dataset for classification through robust Arabic preprocessing and feature engineering. Visualization techniques confirmed that the transformed features contain meaningful structure for fake news detection. These outputs established a strong foundation for model training in Phase 2.

Phase 2 Report: Model Training, Evaluation, and Comparison

1. Introduction

The objective of this phase was to train and compare multiple machine learning and deep learning models for detecting fake news in Egyptian Arabic text.

The following models were evaluated:

1. XGBoost + TF-IDF
2. BiLSTM + Custom Word2Vec
3. AraBERT Transformer

2. Train/Test Split

The cleaned labeled dataset was divided using an 80/20 stratified split:

- Training samples: 5,522
- Testing samples: 1,381

Stratification preserved class balance in both sets.

3. Model 1: XGBoost + TF-IDF

Justification

XGBoost is a powerful gradient boosting model that performs well on sparse text features such as TF-IDF vectors.

Results

- Accuracy: 91.24%
- Fake F1-score: 0.80
- Real F1-score: 0.94

Analysis

The model achieved the highest overall accuracy and strong performance on real news. Some fake samples were misclassified as real due to linguistic similarity.

4. Model 2: BiLSTM + Custom Word2Vec

Justification

BiLSTM captures sequential dependencies in text. Using custom Word2Vec embeddings trained on Egyptian Arabic improves understanding of dialectal vocabulary.

Results

- Accuracy: 90.22%
- Precision: 90.31%
- Recall: 97.73%

Analysis

This model showed the highest recall, indicating strong ability to detect suspicious content while leveraging domain-specific embeddings.

5. Model 3: AraBERT

Justification

AraBERT is a transformer model pretrained specifically for Arabic language tasks.

Results

- Accuracy: 90.01%
- F1-score: 89.87%

Analysis

AraBERT delivered competitive performance but slightly underperformed compared to XGBoost on this short-text dataset.

6. Final Comparison

Model	Accuracy
XGBoost + TF-IDF	91.24%
BiLSTM + Word2Vec	90.22%
AraBERT	90.01%

7. Explainability Analysis (XAI)

To provide deeper insight into the behavior of the best-performing model, Explainable Artificial Intelligence (XAI) techniques were applied to the XGBoost classifier using SHAP (SHapley Additive exPlanations). While evaluation metrics such as accuracy and F1-score indicate how well a model performs, they do not explain *why* a model makes its predictions. Therefore, interpretability methods are important for understanding decision patterns, validating model behavior, and increasing trust in the results.

SHAP is a game-theoretic explanation framework that estimates the contribution of each feature to the final prediction. In this project, SHAP was used to analyze the TF-IDF features learned by XGBoost and determine which lexical signals were most responsible for distinguishing fake news from real news in Egyptian Arabic text.

Global Interpretation

A SHAP beeswarm summary plot was generated to visualize global feature importance across the test dataset. In this plot, features are ranked according to their average influence on predictions. Features appearing near the top of the graph had the greatest effect on the classifier output.

The analysis showed that a limited number of words and tokens had strong positive or negative contributions. Some tokens consistently increased the probability of the fake class, while others were associated with the real class. This indicates that the model successfully identified discriminative lexical cues within the dataset.

The wide spread of SHAP values for several features suggests that these words were highly informative and had varying impact depending on context and frequency. This confirms that not all words contribute equally, and that specific tokens play a dominant role in classification decisions.

Local Interpretation

In addition to global importance, SHAP can also explain individual predictions. For a specific news text or tweet, the method can show which words pushed the model toward predicting fake news and which words pushed it toward predicting real news.

This local interpretability is particularly useful in fake news detection applications, where understanding the reason behind a suspicious classification is often as important as the prediction itself. It can help analysts verify whether the model is focusing on meaningful textual patterns rather than irrelevant noise.

Why XGBoost Outperformed Deep Models

The explainability results help justify why XGBoost slightly outperformed AraBERT and BiLSTM in this project.

- The dataset consists mainly of short tweet-style texts, where exact words and short phrases are often more informative than long-range semantic context.
- TF-IDF preserves sparse lexical patterns and frequency-based signals very effectively.
- XGBoost is highly efficient at learning nonlinear decision boundaries from sparse tabular features.
- With a moderate labeled dataset size (6,903 samples), classical boosting methods may generalize better than deeper neural architectures.
- Deep models such as AraBERT often require larger labeled datasets or additional hyperparameter tuning to fully surpass traditional methods.

These findings indicate that the nature of the dataset strongly favored models that rely on keyword-level evidence.

Practical Importance of Explainability

Explainability is especially valuable in misinformation detection systems because automated decisions may affect public trust and information credibility. A model that can justify its predictions is more useful in real-world settings than a black-box system with no explanation.

By applying SHAP, this project demonstrates that high predictive performance can be combined with transparent decision-making. This strengthens the practical relevance of the proposed fake news detection framework.

Conclusion

SHAP analysis confirmed that the fake news detection task in this dataset depends strongly on lexical cues rather than deep contextual semantics. The XGBoost model successfully leveraged a set of high-impact textual features to achieve the highest classification accuracy.

These results explain why XGBoost performed better than AraBERT and BiLSTM on this specific short-text Egyptian Arabic dataset, and they highlight the importance of combining strong predictive models with interpretable AI methods in NLP applications.

8. Limitations

- Dataset imbalance between real and fake classes
- Limited fake samples compared to real samples
- Short text length reduces context availability

9. Conclusion

Phase 2 demonstrated that both classical and deep learning approaches are effective for Arabic fake news detection. XGBoost with TF-IDF achieved the best overall accuracy, while BiLSTM with custom Word2Vec embeddings showed strong recall and domain adaptation. The project confirms that combining high-quality preprocessing with suitable models can produce reliable fake news detection systems for Egyptian Arabic text.